Examiner only

7. One method of studying the rates of chemical reactions is to use a clock reaction.

One clock reaction involves the reaction between iodate(V) ions, IO_3^-, and hydrogensulfate(IV) ions, HSO_3^-.

A student is given the following three solutions:

- sodium hydrogensulfate(IV) solution containing $HSO_3^-(aq)$
- potassium iodate(V) solution containing $IO_3^-(aq)$
- starch solution

He is also provided with deionised water.

The student combines the following volumes of solutions and measures the time taken for the colour to change.

Volume of HSO_3^- / cm^3	Volume of IO_3^- / cm^3	Volume of starch / cm^3	Volume of deionised water / cm^3	Time / s
10	10	5	25	164
10	20	5	15	82
10	30	5	....	55

(a) Complete the table to show the volume of deionised water that should be used in the final experiment. Give a reason for the value you have chosen. [1]

....

....

(b) Find the order of reaction with respect to iodate(V) ions. Explain how you reached your conclusion. [2]

....

....

(c) The reaction is fourth order overall. Suggest a rate equation for the reaction. [2]

....

(d) Suggest a rate determining step for the reaction. [1]

....

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(e) Addition of lead(II) ions to the mixture after the reaction produces a mixture of products including lead(II) iodide and lead(II) iodate(V).

(i) Give the colour of lead(II) iodide. [1]

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(ii) A sample of lead(II) iodate(V) decomposes on heating to give a mixture of solids and a mixture of oxygen gas and iodine vapour.

At a temperature of 200 °C, the volume of gas is 220 cm^3, but when cooled to 20 °C the iodine solidifies and the volume of gas is 110 cm^3.

Calculate the number of moles of gas at these two temperatures and hence find the percentage of iodine molecules in the original gas mixture. [3]

[All volumes are measured at a pressure of 1.01×10^5 Pa]

Number of moles of gas at 200 °C = ... mol

Number of moles of gas at 20 °C = ... mol

Percentage of iodine molecules = ... %

10

